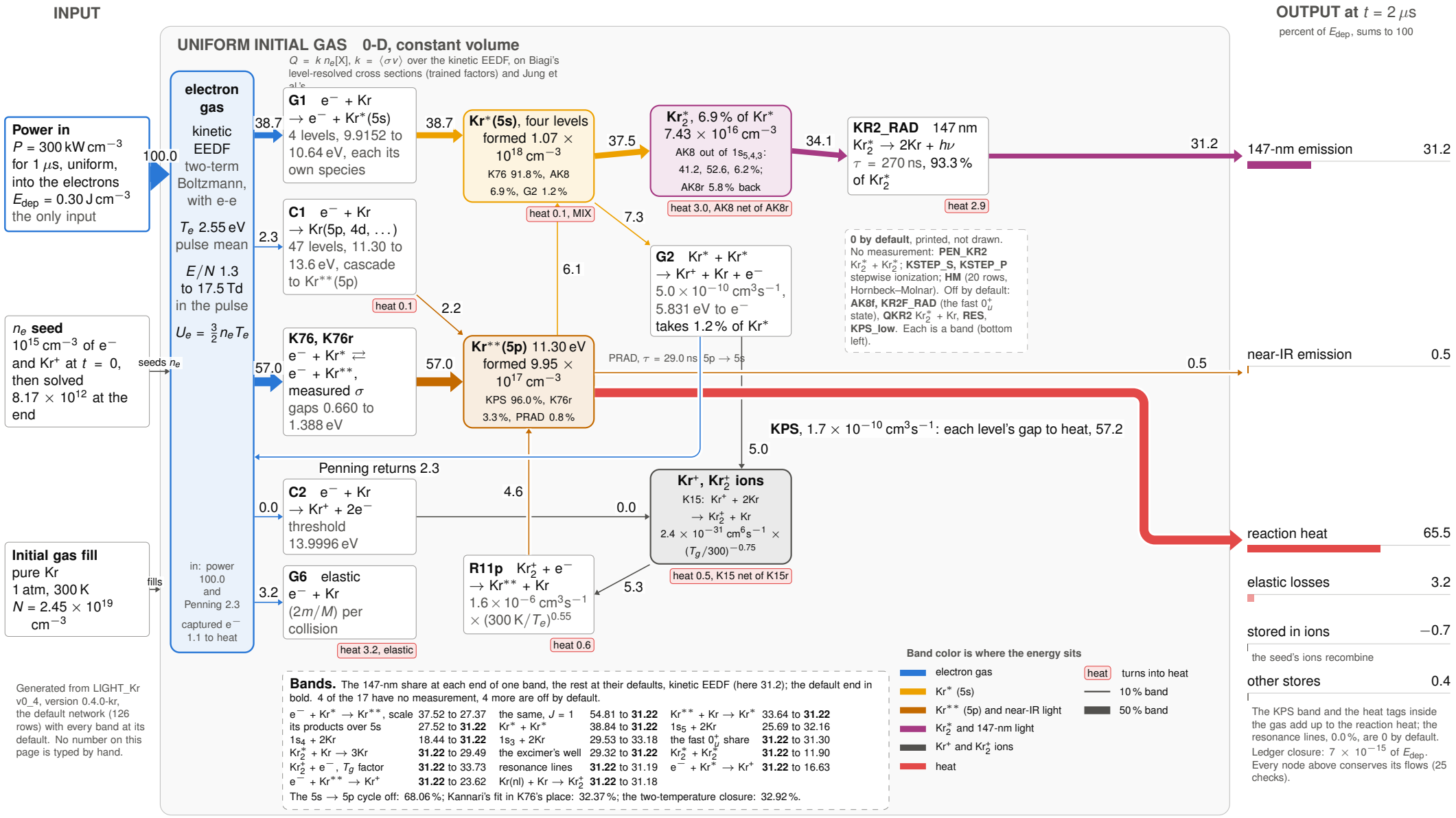
LIGHT_Kr system map, v0_4: energy flow through pure krypton

Reference point: 1 atm of Kr at 300 K, P = 300 kW cm^{-3} into the electrons for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e seeded at 10^{15} cm^{-3} and then solved, every band at its default, run to 2 μs on the kinetic EEDF. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1. The four 5s levels are one node, Kr*, each at its own energy.



The outputs as flows, which overlap: to Kr* and Kr** by electron impact 41.0%, to Kr+ 5.0%, to Kr$_2^*$ 33.5% (formed times the 147-nm quantum), heat 65.5%, elastic losses 3.2%. **The 5s ↔ 5p cycle.** The electrons pump the 5s levels up to the 5p rung (K76, on Jung et al.'s measured cross sections) 14 times for each Kr$_2^*$ formed net, and krypton relaxes the rung back onto every level (KPS), so the cycle is drawn as its two net bands: of the 5s atoms formed (gross), 91.8% are pumped up; the rung relaxes 96.0% of it on Kr, radiates 0.8% and the electrons return 3.3%; each level's gap of every net cycle, 9.51 × 10^{17} cm^{-3} of them, takes 57.0% of the input from the electrons, and KPS turns it into heat. The 5s atoms come from G1 (6.7%), the rung's KPS (89.1%) and PRAD (0.7%), K76r (3.0%) and the excimer's redissociation (0.4%). Penning ionization (G2) makes 100% of the ions, and they recombine onto the rung (R11p). **The kinetic EEDF:** 348 solutions, 274 in the pulse; over its second half the field carries 1.0004 to 1.0007 of the input and the kinetic and the model's energy balances agree to 7 × 10^{-6} of it. At the end T_g = 707 K, p = 2.36 atm, n_e = 8.17 × 10^{12} cm^{-3}. **Sources** (DOIs) of the rates printed: cross sections Biagi 1999, 10.1016/S0168-9002(98)01233-9, via LXCat, 10.1016/j.chemphys.2011.04.020, times factors trained on krypton's drift velocities and ionization coefficients (5s ×0.89, 5p ×0.92, higher ×0.75, momentum transfer ×0.93 to ×1.23); K76 Jung et al. 2006, 10.1103/PhysRevA.73.022722; AK8 Marchal et al. 2018, 10.5772/intechopen.71942, on Wieme and Lenaerts 1981, 10.1063/1.440855; AK8r LaRocque et al. 1986, 10.1063/1.450716, Killeen and Eden 1986, 10.1063/1.450794; MIX Turner 1967, 10.1103/PhysRev.158.121, Audouard et al. 1991, 10.1063/1.460834; KR2_RAD Morikawa et al. 1989, 10.1063/1.457108; G2 Leichner and Ericson 1974, 10.1103/PhysRevA.9.251, as Kannari et al. 1985 print it, 10.1063/1.334590; KPS Gazeli et al. 2021, 10.1063/5.0041471; PRAD Dzierżęga et al. 2000, 10.1103/PhysRevA.62.022505, and NIST ASD, 10.18434/T4W30F; R11p Shiu and Biondi 1977, 10.1103/PhysRevA.16.1817; K15 Bhattacharya 1970, 10.1103/PhysRevA.1.1196, Grössl et al. 1981, 10.1063/1.441261, Smith et al. 1972, 10.1088/0022-3700/5/11/020; K15r Signorell et al. 2001, 10.1063/1.1370939.